Artificial Intelligence Project Report

Project Title: Data analysis on Bank Marketing dataset using machine learning

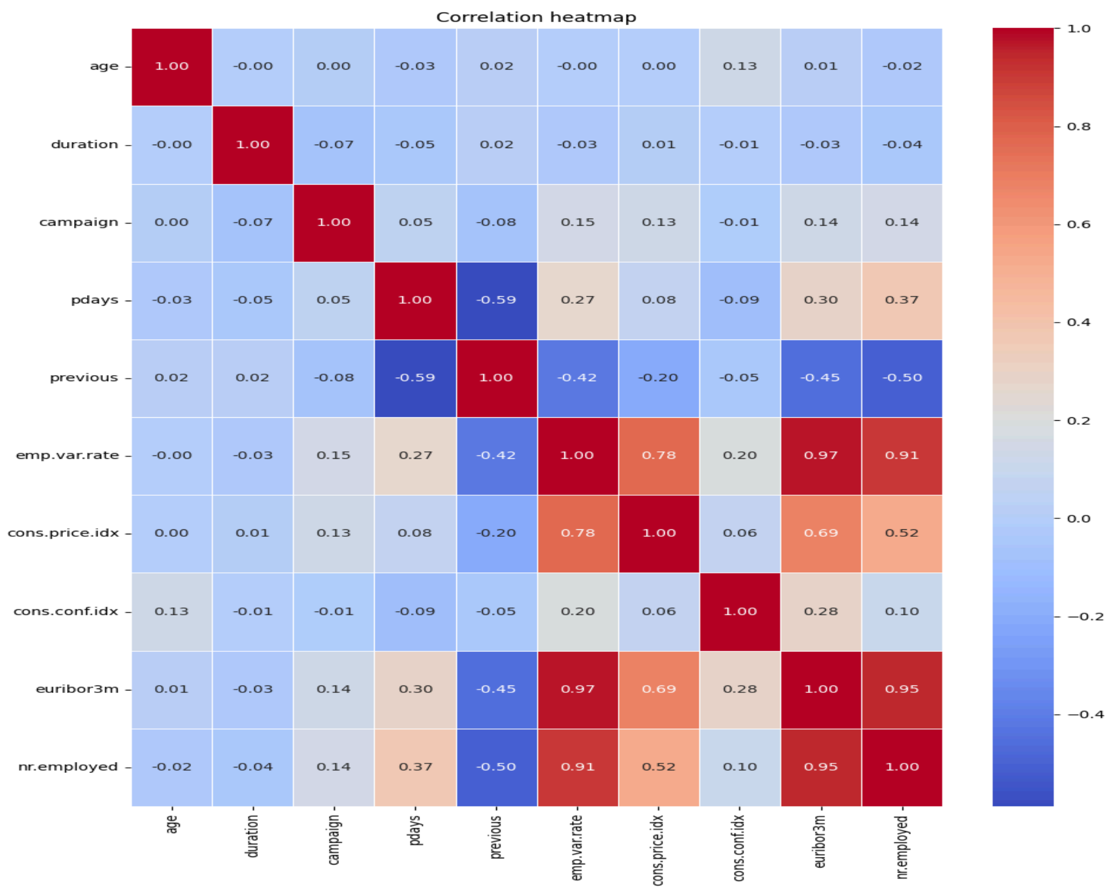
Table of Contents

1. Introduction

This project aims to create a predictive model that can predict if a client will sign up for a bank term deposit based on how they interact with a series of marketing campaigns. We will use this "Bank Marketing" dataset, which has 20 features and 41,188 datapoints, to train and test our model. The main goal of the project is to use different machine learning models to make better predictions about the target variable and compare the accuracy of the models at the same time.

2. Dataset description

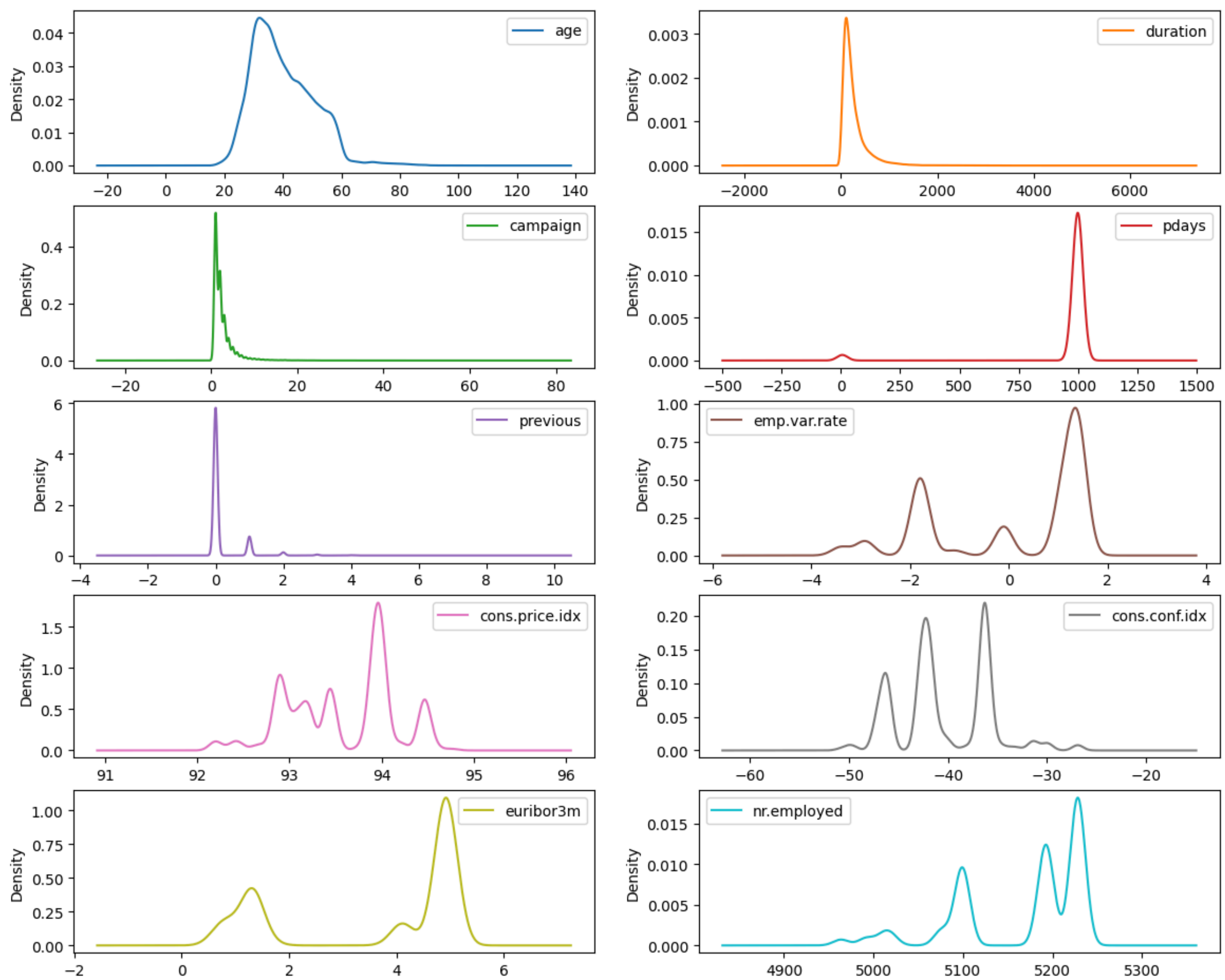
Dataset Description: The “Bank Marketing” dataset having 20 features and 41,188 data points is a binary classification problem since in the target column ‘y’, there are only two classes either ‘yes’ or ‘no’. Moreover, in this dataset, there are both quantitative and categorical features. The quantitative features include the columns: age, duration, campaign, pdays, previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m and, nr.employed. On the other hand, the categorical features include the columns: job, marital, education, default, housing, loan, contact, month, day_of_week, poutcome, and y. As there are categorical variables in our dataset, we had to encode them. Because while training a machine learning model, we cannot use string values directly to train the model. So, first we need to encode them. As our target column is a binary class, we used label encoding.



From the correlation heatmap, we can see that the features like emp.var.rate, euribor3m, and nr.employed have high positive correlation (> 0.90). On the other hand, variables such as pdays and previous have moderate negative correlation (-0.59).

Imbalanced dataset: The dataset is imbalanced as in our target column 'y' the count of no is 36,548 which consists 88.73% and the count of yes is 4,640 which consists 11.27% of total dataset.

Density plot of Numerical features



3. Dataset pre-processing

Missing values: First, when we checked for missing values or null values, it showed 0 for all columns, though it contained ‘unknown.’ After checking the count of ‘unknown,’ we found that there are some unknown values in some of the features. After replacing ‘unknown’ with null, again we checked for the missing values, and then it showed the count of missing values. After this we used imputation to fill the missing values by mean values.

Categorical Values: For categorical variables, we used label encoding.

Feature Scaling: We used standard scaling for the quantitative variables.

Dropping column: We dropped the duration column because it causes data leakage, as the call length is only known after the outcome is already decided. Including it would result in a "leaky" model that appears highly accurate in testing but fails to predict subscriptions in real-world scenarios where the duration is unknown beforehand. This ensures the model relies on more meaningful predictors like economic indicators and client demographics.

4. Dataset splitting

For dataset splitting, we used 80% data for model training and 20% data for model testing. As our dataset was imbalanced, we used stratified sampling to ensure the splitting is done in a balanced way.

5. Model training & testing (Supervised)

For model training and testing, we used Logistic Regression, KNN, Naive Bayes, and Neural Network(MLP).

1. Logistic Regression: Logistic Regression was implemented as a baseline model because it is highly efficient for binary classification tasks. It provides a clear probabilistic interpretation of how features like age or interest rates directly influence the likelihood of a client subscribing to a term deposit.

2. k-Nearest Neighbors (k-NN): This model was selected because it is non-parametric and classifies data points based on their similarity to neighboring instances. It is particularly useful for identifying local patterns in customer behavior that might not be captured by linear boundaries.

3. Naive Bayes: Naive Bayes was used due to its computational efficiency and its ability to handle categorical features effectively. Despite its assumption of feature independence, it serves as a robust probabilistic classifier that often performs surprisingly well on marketing datasets.

4. Neural Network (Multilayer Perceptron): A Neural Network was included to capture complex, non-linear relationships between the 20 different features in the dataset. Its multi-layer architecture allows the model to learn deep patterns and interactions that simpler algorithms might overlook, potentially leading to higher predictive accuracy.

6. Model selection/ Comparison analysis

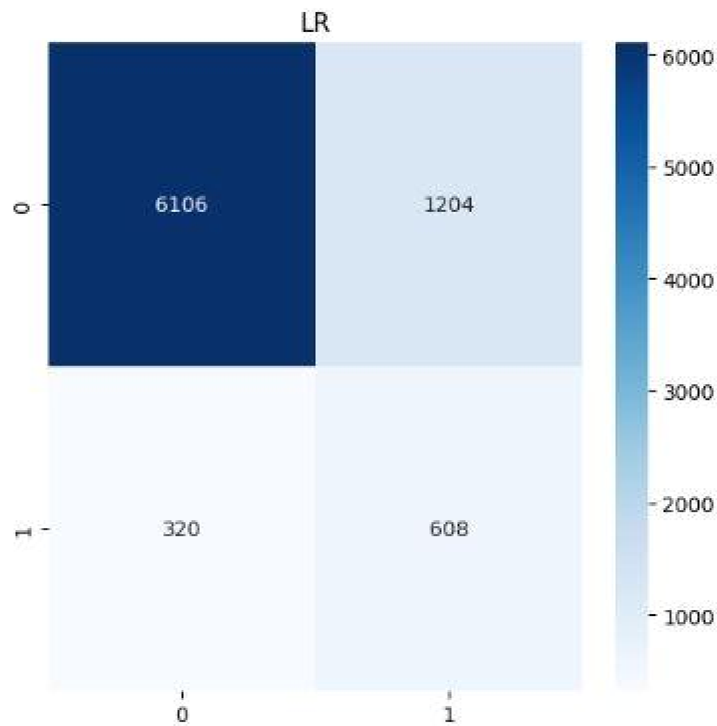
1. Logistic Regression:

- Accuracy: 81.50%
- Precision Score: 0.336
- Recall Score: 0.655
- F1 Score: 0.444

```
Logistic Regression Accuracy: 0.8150036416605972
              precision    recall  f1-score   support

           0       0.95      0.84      0.89      7310
           1       0.34      0.66      0.44       928

    accuracy                           0.82      8238
   macro avg       0.64      0.75      0.67      8238
weighted avg       0.88      0.82      0.84      8238
```



Analysis: This model provides a relatively balanced performance between identifying subscribers and maintaining a decent accuracy level. It has the second-highest F1-score, suggesting it is more reliable than others for general targeting.

2. k-Nearest Neighbors (k-NN):

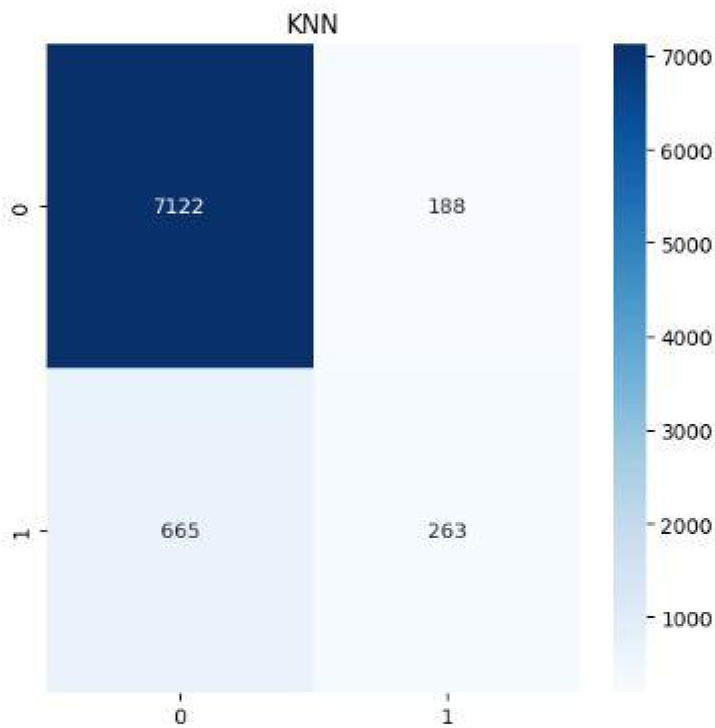
- Accuracy: 89.65%

- Precision Score: 0.583
- Recall Score: 0.283
- F1 Score: 0.381

```
KNN Accuracy: 0.8964554503520272
              precision    recall  f1-score   support

           0       0.91      0.97      0.94      7310
           1       0.58      0.28      0.38       928

    accuracy                           0.90      8238
   macro avg       0.75      0.63      0.66      8238
weighted avg       0.88      0.90      0.88      8238
```



Analysis: k-NN achieves high accuracy by heavily favoring the majority class, resulting in poor recall for actual subscribers. While it is precise when it predicts a "yes," it misses over 70% of potential customers.

3. Naive Bayes:

- Accuracy: 19.42%

- Precision Score: 0.120
- Recall Score: 0.969
- F1 Score: 0.213

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Naive Bayes Accuracy: 0.19422189851905802
              precision    recall  f1-score   support

           0       0.96      0.10      0.17      7310
           1       0.12      0.97      0.21       928

    accuracy                           0.19      8238
   macro avg       0.54      0.53      0.19      8238
weighted avg       0.87      0.19      0.18      8238
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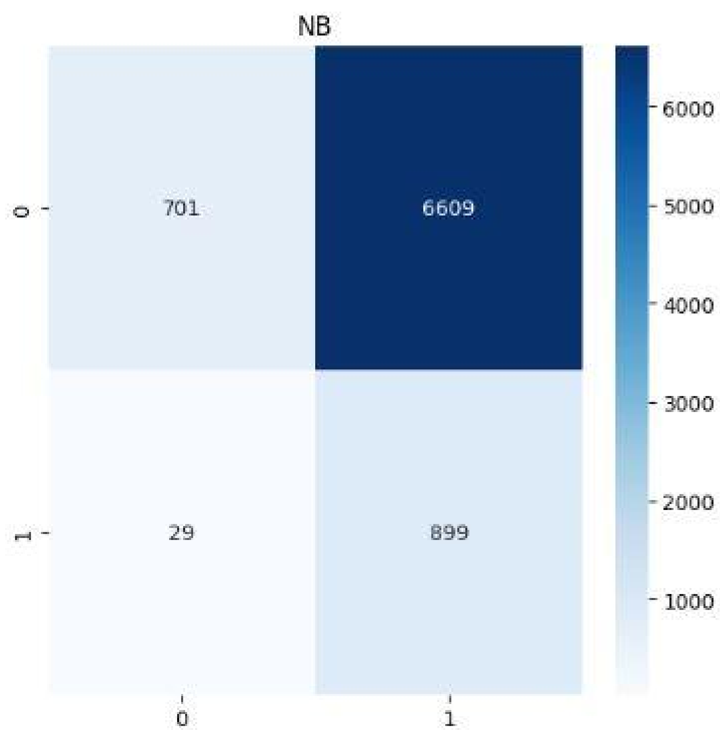


Analysis: This model has extremely low accuracy because it predicts almost every client will subscribe. While the recall is nearly perfect, the precision is very low, meaning the bank would waste resources contacting thousands of people who will not subscribe.

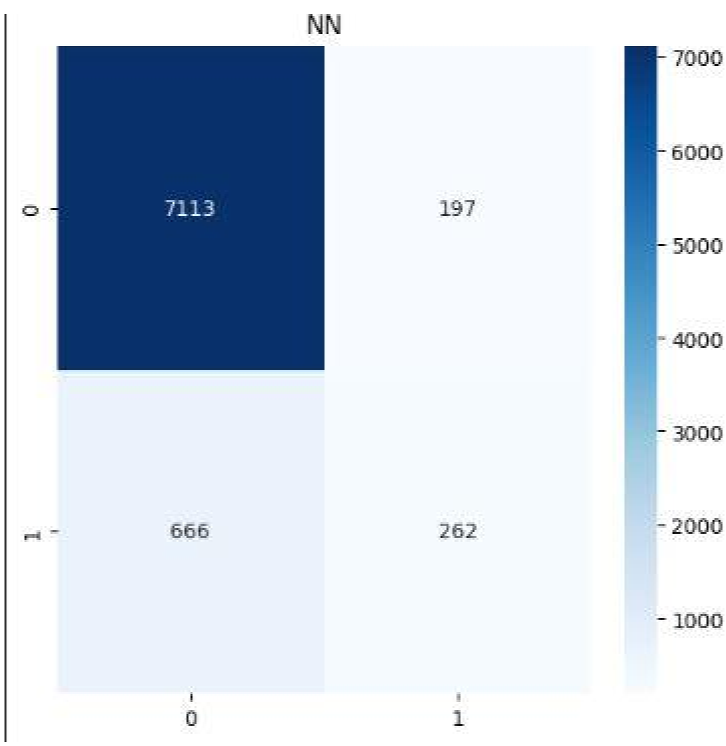
4. Neural Network (MLP):

- Accuracy: 89.52%
- Precision Score: 0.571
- Recall Score: 0.282
- F1 Score: 0.378

```
Neural Network Accuracy: 0.895241563486283
              precision    recall  f1-score   support

           0       0.91      0.97      0.94      7310
           1       0.57      0.28      0.38       928

    accuracy                           0.90      8238
   macro avg       0.74      0.63      0.66      8238
weighted avg       0.88      0.90      0.88      8238
```



Analysis: The Neural Network performs very similarly to k-NN, offering high accuracy but struggling to correctly identify the minority "yes" class. It is conservative in its predictions, leading to low recall and a moderate F1-score.

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AUC Scores:
Logistic Regression: 0.7955
KNN: 0.7258
Naive Bayes: 0.7710
Neural Network: 0.7559
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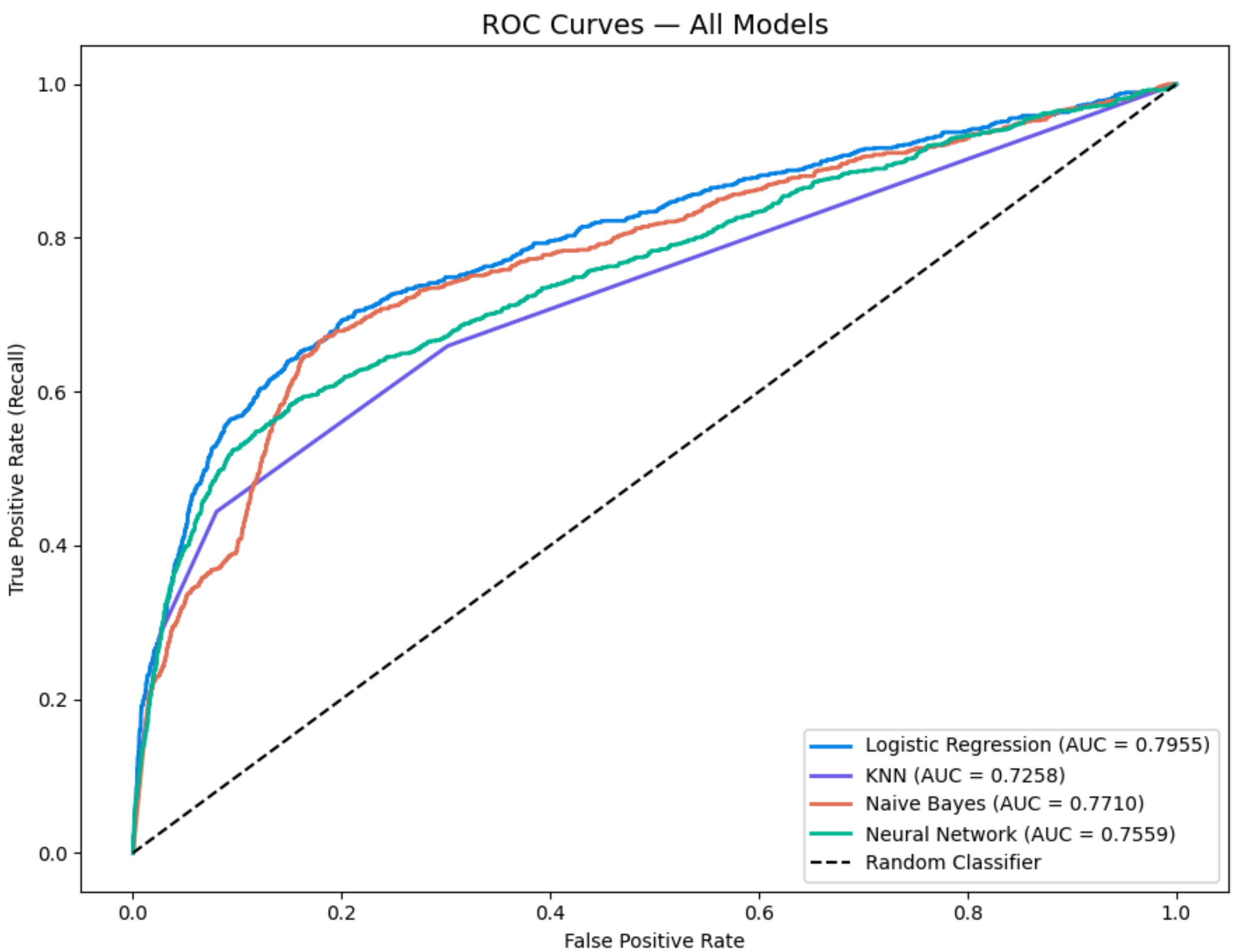


7. Conclusion

Understanding the Results:

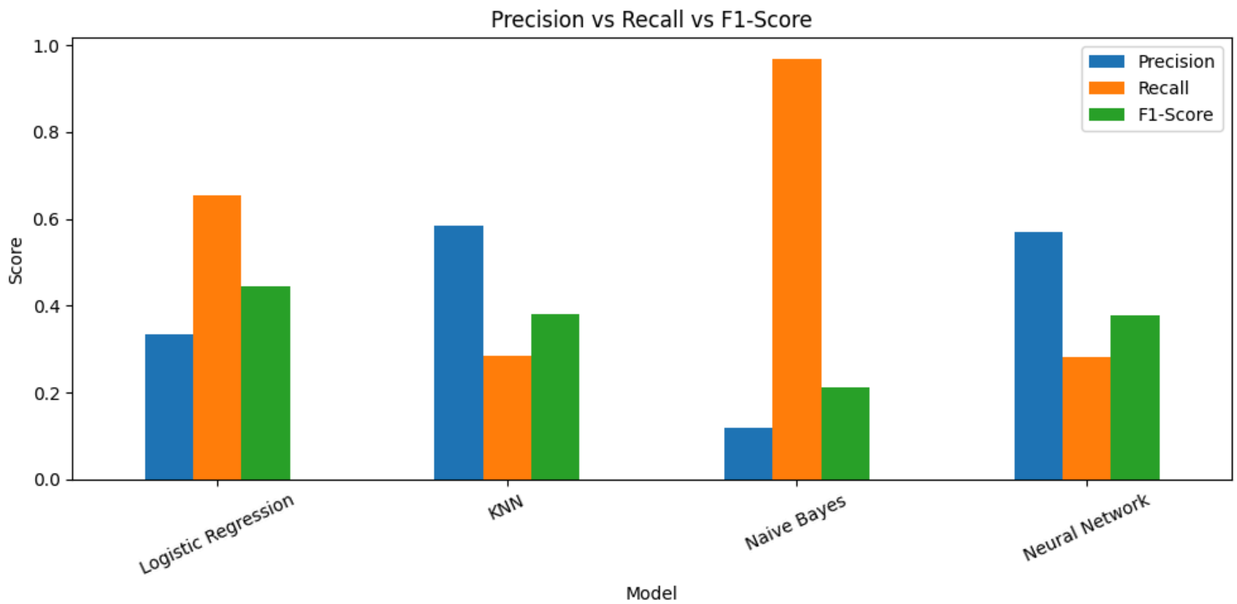
Based on the analysis of the Bank Marketing dataset, several key insights emerge regarding the factors that drive term deposit subscriptions. The dataset, consisting of over 41,000 entries, reveals a target audience primarily centered around a median age of 40 years. There is a strong relationship between social and economic indicators—such as the Euribor 3-month rate and employment variation rates—and the success of marketing campaigns.

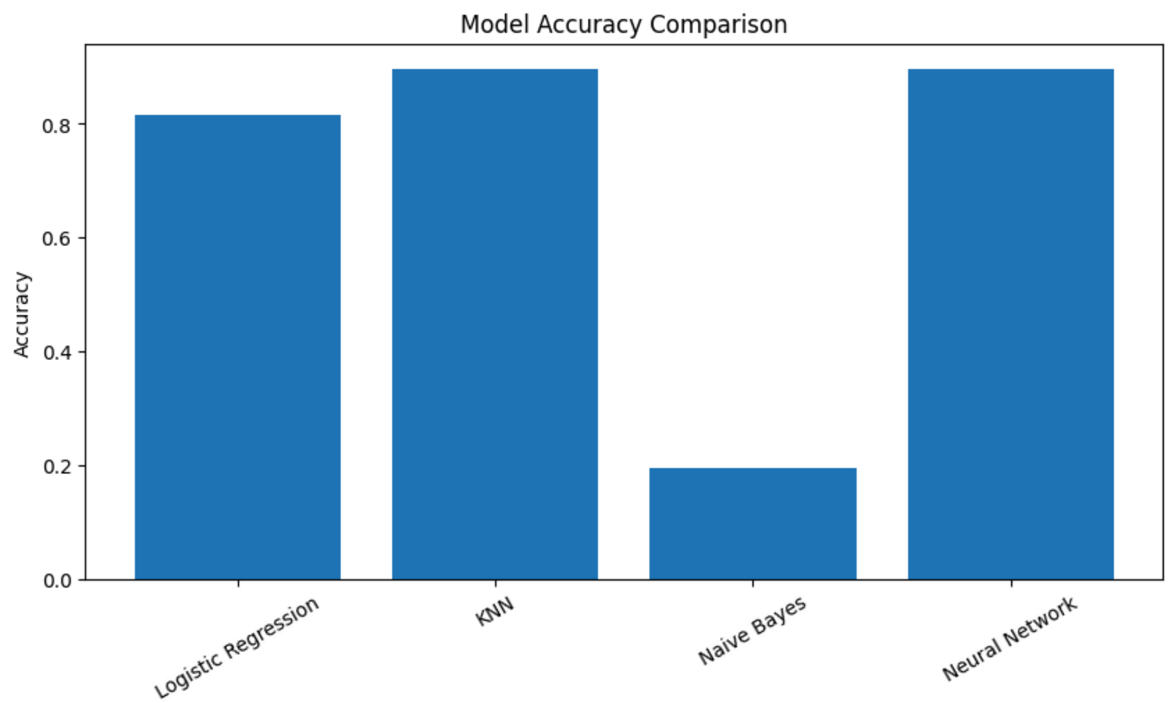
Performance Comments:

The comparison of the four models (Logistic Regression, KNN, Naive Bayes, and Neural Network) shows a significant trade-off between accuracy and reliability:

- KNN and Neural Networks achieved the highest overall accuracy (reaching approximately 90%), making them the most effective at general prediction.
- Naive Bayes displayed a very high Recall (near 1.0) but suffered from extremely low Precision and Accuracy (around 20%). This suggests the model is over-predicting the "yes" class, which leads to many false positives.
- Logistic Regression offered a more balanced performance with an F1-Score near 0.45, suggesting it is more stable than Naive Bayes despite having lower accuracy than the Neural Network.

Model	Precision	Recall	F1-Score
Logistic Regression	0.335541	0.655172	0.443796
KNN	0.583149	0.283405	0.381436
Naive Bayes	0.119739	0.968750	0.213134
Neural Network	0.570806	0.282328	0.377794





Reason behind getting such result:

The high accuracy of KNN and Neural Networks likely stems from their ability to capture complex, non-linear relationships within the 21 features provided. However, the extremely high recall in Naive Bayes indicates a struggle with the class imbalance inherent in marketing data—where "no" outcomes significantly outnumber "yes" outcomes. Additionally, the feature duration heavily influences results; while it is a strong predictor, it is only available after a call ends, which can lead to overly optimistic performance metrics in a real-world predictive setting.

Challenges Faced:

- Handling Imbalanced Data: A major hurdle was ensuring the models didn't simply default to predicting "no" to achieve high accuracy.
- Feature Complexity: Managing 11 categorical variables required careful encoding to maintain data integrity without creating an excessively sparse feature matrix.
- Model Selection: Finding a balance between a model that catches all potential subscribers (High Recall) versus one that is actually correct when it predicts a subscription (High Precision) was the central technical challenge of this project.